Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

MT407 – MECHANICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define Tension.
2. Define Equilibrium of a particle.
3. Define angle of friction.
4. What is triangular lamina?
5. Define Resultant velocity.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. The magnitude of the resultant of two given forces P, Q is R. If Q is doubled then R is doubled. If Q is reversed then also R is doubled. Show that P:Q:R= $\sqrt{2} : \sqrt{3} : \sqrt{2}$.
7. If three forces keep a particle in equilibrium then prove that the three forces are coplanar.
8. If forces of magnitude 3P,4P,5P act along the sides BC,CA,AB of an equilateral triangle of side a. Find the moment of the resultant about A.
9. Prove that the mass centre of three uniform rods forming a triangular frame is the in-centre of the triangle.
10. A particle has two velocities v$\overrightarrow{V_1}$ 1 and v2. Its resultant velocity is equal to v1 in magnitude. Show that when the velocity v1 is doubled the new resultant is perpendicular to v2.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Find the magnitude and direction of the resultant of F1 and F2.

(or)

b) State and prove Lami's theorem.

12. a) Find the resultant of two parallel forces acting on a rigid body.

(or)

b) Three forces acting along the sides of a triangle in the same order are equivalent to a couple. Show that they are proportional to the sides of the triangle.

13. a) Suppose a particle of weight W lying on a rough plane inclined at an angle α to the horizontal is subject to a force P along the plane in the upward direction. If the equilibrium is limiting. Find the value of P.

(or)

b) A particle rests on a plane inclined at 45° to the horizontal being supported by a string along the line of the greatest slope. If the ratio of the maximum and minimum tensions consistent with equilibrium is 2:1. Find the coefficient of friction. V

14. a) If D, E, F the midpoints of the sides BC, CA, AB of a triangular lamina ABC. The lamina is folded across the line joining E, F and the vertex A is made to lie on the base BC. Show that the distance of the mass centre of the lamina in this position from BC is three fourths of the distance of the mass centre of the unfolded lamina from BC.

(or)

b) Find the mass centre of solid hemisphere of radius a?

15. a) Find the magnitude and direction of the resultant of the velocities $\overrightarrow{v_1}$ and $\overrightarrow{v_2}$

(or)

b) Find the components of velocity and acceleration of a particle in the radial and transverse directions?

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